Source: Switching Power Supply Design

CHAPTER 4
Flyback Converter Topologies

Foreword

I find that many engineers and students have great difficulty with the design of flyback type converters. This is unfortunate because these topologies are very useful, and, in fact, they are not difficult to design.

The problem is not the intrinsic difficulty of the subject matter (or the ability of the student). The fault is related to the way the subject is traditionally taught.

Right from the start the normal term *flyback transformer* immediately projects the wrong mindset. Not unreasonably, designers set out to design a "flyback transformer" as if it were a real transformer. This is not the way to go.

We are all very familiar with transformers, very simple devices really—we put a voltage across a primary winding and we get a voltage on a secondary winding. The voltage ratio follows the turns ratio, irrespective of the output (or load) current. In other words, the transformer conserves the voltage transfer ratio (one volt per turn on the primary results in one volt per turn on the secondary. You want ten volts? Then use ten turns, very simple). However, notice an important property of transformers, the primary and secondary conduct at the same time. If current flows into the start of the primary winding it flows out of the start of the secondary winding at the same time.

Figure 4.1 shows the basic schematic of a flyback converter. Notice the when $Q1$ is "on" current flows into the primary winding of $T1$ but the secondary diodes are not conducting and there is no secondary current. When $Q1$ turns "off" the primary current stops, all winding voltages reverse by flyback action, and the output diodes and secondary windings now conduct current. So the primary and secondary windings in the flyback "transformer" conduct current at different times. This apparently minor difference dramatically changes the rules.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

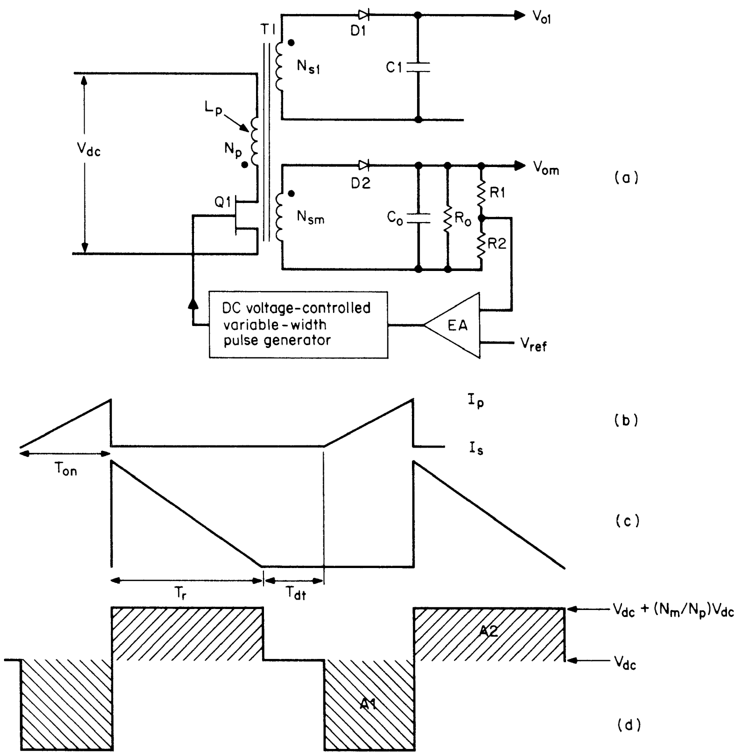


FIGURE 4.1 Basic flyback converter schematic. The action is as follows: When $Q1$ turns "on," all rectifier diodes become reverse-biased, and all output load currents are supplied from the output capacitors. T1 acts like a pure inductor and primary current builds up linearly in it to a peak I_p. When $Q1$ turns "off," all winding voltages reverse under flyback action, bringing the output diodes into conduction and the primary stored energy $1/2LI_p^2$ is delivered to the output to supply load current and replenish the charge on the output capacitors (the charge that they lost when $Q1$ was on). The circuit is discontinuous if the secondary current has decayed to zero before the start of the next turn "on" period of $Q1$.

Think about it! When $Q1$ is "on" only the primary winding is conducting (the other windings are not visible to the primary because they are not conducting). $Q1$ thinks it is driving an inductor. When $Q1$ turns "off" only the secondary windings conduct and now the primary winding cannot be seen by the secondaries (so now the secondaries think they are being driven by an inductor). So how does this change the rules? Well, functionally the so-called flyback "transformer" is really functioning as an inductor with several windings and follows the rules applicable to inductors.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

The rules for an inductor with more than one winding are as follows: The primary to secondary ampere-turns ratios are conserved (not the voltage ratios, as was the case with a true transformer). For example, if the primary is, say, 100 turns and the current when $Q1$ turns "off" is 1 amp, then we have developed 100 ampere-turns in the primary. This must be conserved in the secondaries. With, say, a single secondary winding of 10 turns, the secondary current will be 10 amps (10T × 10A = 100 ampere-turns). In the same way, a single turn will develop 100 amps or 1000 secondary turns will develop 0.1 amps.

So where do we stand with regard to voltage? Well, to the first order, there is no correlation between primary and secondary voltages. The secondary voltage is simply a function of load. Consider the 10-turn 10-amp (100 ampere-turns) secondary winding example mentioned above. If we terminate the winding with a 1-ohm load, we will get 10 volts. What is more striking because the 10 amps must be conserved is that if we terminate it with 100 ohms, we will get 1000 volts! This is why the flyback topology is so useful for generating high voltages (don't try to open circuit this winding because it will destroy the semiconductors). With several secondary windings conducting at the same time, then the sum of all the secondary ampere-turns must be conserved.

So the lesson we learn here is that flyback "transformers" actually operate as inductors and must be designed as such. (In Chapter 7, I use the term *choke* instead of inductor because the core must support both DC and AC components of current.) If flyback "transformers" had originally been called by their correct functional name, "flyback chokes," then a lot of confusion could have been avoided.

We must not forget that voltage transformation is still taking place between primary and secondary windings even if they are not conducting at the same time. Taking the above example of 10 turns terminated in 100 ohms, the 1000 volts thus developed on this secondary winding will reflect back to the primary as 10,000 volts; this added to the supply of 100 volts will stress $Q1$ in its "off" state with 10,100 volts (where did I put that 11,000 volt transistor?). Hardly practical, but the theory holds.

So when designing flyback transformers keep the following key points in mind:

1. Remember you are not designing a transformer, you are designing a choke with additional windings.
2. The primary turns are selected to satisfy the AC voltage stress (volt-seconds) and the core AC saturation properties:

$$N_p = \frac{VT}{B\,Ae}$$

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Where Np is minimum primary turns
V is the maximum primary DC voltage (volts)
T is the maximum "on" period for $Q1$ (microseconds)
B is the AC p-p flux swing (tesla) typically 200 mT for ferrite
Ae is the effective center pole area of the core (mm^2)

3. The secondary turns are optional. If you choose the same volts per turn on the secondary as was used for the primary, then the flyback voltage on $Q1$ will be twice the supply voltage.
4. When using a gapped ferrite core, the minimum core gap must be such that the core will not saturate for the sum of DC and AC magnetization current. More often the gap is chosen to satisfy the power transfer requirements. This normally results in a gap exceeding the minimum requirements. Remember the energy stored in the primary is

$$E(joules) = {}^1/_2 LI^2$$

Remember this is the maximum energy that can be transferred to the secondary, and then only in the discontinuous (complete energy transfer) mode. In the continuous mode, only part of this energy is transferred.

Note *Although reducing the inductance* L *may appear to reduce the stored energy, the current* I *increases in the same ratio as the inductance decreases. Since the* I *parameter is squared, the stored energy actually increases as* L *decreases.*

5. It is not recommended that you try to design for a defined inductance. It is better to let inductance be a dependant variable as changing the core gap or core material (permeability) will change the inductance.

Below in Chapter 4, Pressman follows the conventional "flyback transformer" approach, providing a very complete analysis. The reader may find it helpful to first read Chapter 7 in this book and Part 2, Chapters 1 and 2 in my book, shown as Reference 1 at the end of this chapter.

4.1 Introduction

All the topologies previously discussed (with the exception of the boost regulator Section 1.4 and the polarity inverter Section 1.5) deliver power to their loads during the period when the power transistor is turned "on."

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

However, the flyback topologies described in this chapter operate in a fundamentally different way. During the power transistor "on" time, they store energy in the power transformer. During this period, the load current is supplied from an output filter capacity only. When the power transistor turns "off," the energy stored in the power transformer is transferred to the load and to the output filter capacitor as it replaces the charge it lost when it alone was delivering load current.

The flyback has advantages and limitations, discussed in more detail later. A major advantage is that the output filter inductors normally required for all forward topologies are not required for flyback topologies because the transformer serves both functions. This is particularly valuable in low-cost multiple output power supplies yielding a significant saving in cost and space.

4.2 Basic Flyback Converter Schematic

The basic flyback converter topology together with typical current and voltage waveforms is shown in Figure 4.1. It is very widely used for low-cost applications in the power range from about 150 W down to less than 5 W. Its great initial attraction is immediately clear—it has no secondary output inductor, and the consequent saving in cost and volume is a significant advantage.

In Figure 4.1, flyback operation can be easily recognized from the position of the dots on the transformer primary and secondary (these dots show the starts of the windings). When *Q*1 is "on," the dot ends of all windings are negative with respect to their no-dot ends. Output rectifier diodes *D*1 and *D*2 are reverse-biased and all the output load currents are supplied from storage filter capacitors *C*1 and *C*2. These will be chosen as described below to deliver the load currents with the maximum specified ripple or droop in output voltages.

4.3 Operating Modes

There are two distinctly different operating modes for flyback converters: the continuous mode and the discontinuous mode. The waveforms, performance, and transfer functions are quite different for the two modes, and typical waveforms are shown in Figure 4.2. The value of primary inductance and the load current determine the mode of operation.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

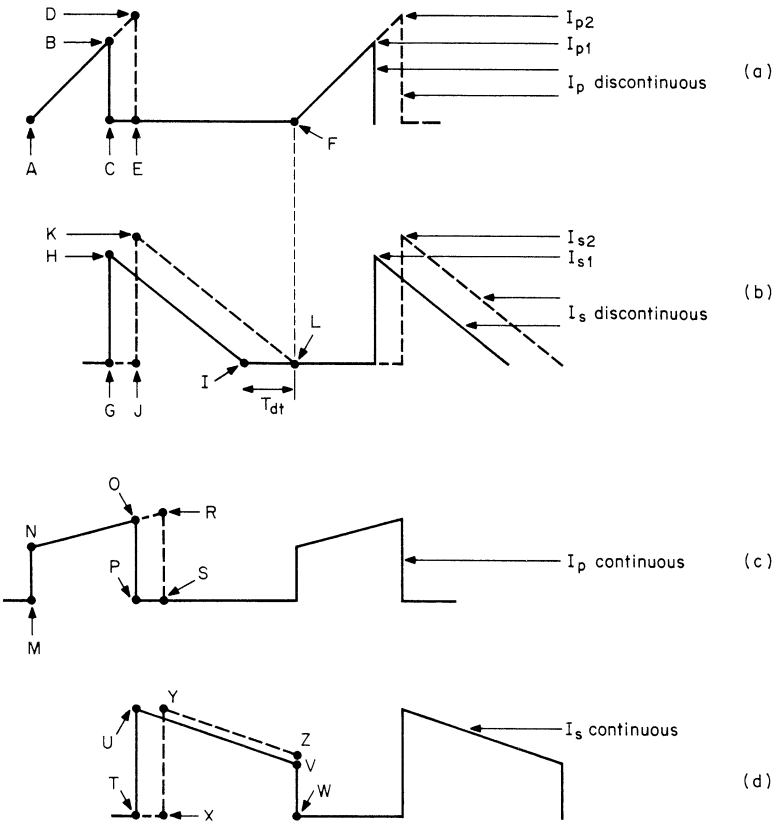


FIGURE 4.2 (*a* and *b*) Waveforms of a discontinuous-mode flyback at the point of transition to continuous-mode operation. Notice in the discontinuous mode, the current remains discontinuous (the transformer has periods of zero current) providing there is a dead time (T_{dt}) between the instant the secondary current reaches zero and the start of the next "on" period. (*c* and *d*) If the transformer is loaded beyond this point, some current remains in the transformer at the end of the "off" period and the next "on" period will have a sharp current step at its front end. This step is characteristic of the continuous mode of operation, as the secondary current no longer decays to zero at any part of the conduction period. There is a dramatic change in the transfer function at the point of entering continuous mode, and if the error-amplifier bandwidth has not been drastically reduced, the circuit will oscillate.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

4.4 Discontinuous-Mode Operation

Figure 4.1 shows a master output and one slave output. As in all other topologies shown previously, a negative-feedback loop will be closed around the master output V_{om}. A fraction of V_{om} will be compared to a reference, and the error signal will control the "on" time of $Q1$ (the pulse width), so as to regulate the sampled output voltage equal to the reference voltage against line and load changes. Hence, the master output is fully regulated. However, the slaves will also be well regulated against line changes and somewhat less well against load changes because the secondary winding voltages tend to track the master voltage. As a result, the slave line and load regulation is better than for the previously discussed forward-type topologies.

During the $Q1$ "on" time, there is a fixed voltage across N_p and current in it ramps up linearly (Figure 4.1*b*) at a rate of $dI/dt = (V_{\mathrm{dc}}-1)/L_p$, where L_p is the primary magnetizing inductance. At the end of the "on" time, the primary current has ramped up to $I_p = (V_{\mathrm{dc}}-1)T_{\mathrm{on}}/L_p$. This current represents a stored energy of

$$E = \frac{L_p(I_p)^2}{2} \tag{4.1}$$

where E is in joules
L_p is in henries
I_p is in amperes

Now when $Q1$ turns "off," the current in the magnetizing inductance forces a reversal of polarities on all windings. (This is called *flyback action*.) Assume, for the moment, that there are no slave windings and only the master secondary N_m. Since the current in an inductor cannot change instantaneously, at the instant of turn "off," the primary current transfers to the secondary at an amplitude $I_s = I_p(N_p/N_m)$.

After a number of cycles, the secondary DC voltage has built up to a magnitude (calculated below) of V_{om}. Now with $Q1$ "off," the dot end of N_m is positive with respect to its no-dot end and current flows out of it, but ramps down linearly (Figure 4.1*c*) at a rate $dI_s/dt = V_{\mathrm{om}}/V_s$, where L_s is the secondary inductance. The discontinuous mode action is defined as follows.

If the secondary current has ramped down to zero before the start of the next $Q1$ "on" time, all the energy stored in the primary when $Q1$ was "on" has been delivered to the load and the circuit is said to be operating in the discontinuous mode.

Since an amount of energy E in joules delivered in a time T in seconds represents input power in watts, we can calculate the input

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

power as follows: At the end of one period, power P drawn from V_{dc} is

$$P = \frac{{}^1\!/_2 L_p (I_p)^2}{T} W \tag{4.2a}$$

But $I_p = (V_{dc} - 1)T_{on}/L_p$. Then

$$P = \frac{[(V_{dc} - 1)T_{on}]^2}{2TL_p} \approx \frac{(V_{dc}T_{on})^2}{2TL_p} W \tag{4.2b}$$

As can be seen from Eq. 4.2*b*, the feedback loop maintains constant output voltage by keeping the product $V_{dc}T_{on}$ constant.

4.4.1 Relationship Between Output Voltage, Input Voltage, "On" Time, and Output Load

Let us assume an efficiency of 80%, then

$$\begin{aligned}\text{Input power} &= 1.25\ (\text{output power})\\ &= \frac{1.25(V_o)^2}{R_o} = \frac{{}^1\!/_2(L_p I_p^2)}{T}\end{aligned}$$

But $I_p = \underline{V_{dc}}\overline{T_{on}}/L_p$ since maximum "on" time $\overline{T_{on}}$ occurs at minimum supply voltage $\underline{V_{dc}}$, as can be seen from Eq. 4.2*b*.

Then $1.25\,(V_o)^2/R_o = {}^1\!/_2 L_p \underline{V_{dc}}^2\overline{T_{on}^2}/L_p^2 T$ or

$$V_o = \underline{V_{dc}}\,\overline{T_{on}}\sqrt{\frac{R_o}{2.5TL_p}} \tag{4.3}$$

Thus the feedback loop will regulate the output by decreasing T_{on} as V_{dc} or R_o goes up, increasing T_{on} as $V_{dc}R_o$ goes down.

4.4.2 Discontinuous-Mode to Continuous-Mode Transition

In Figures 4.2*a* and 4.2*b* the solid lines represent primary and secondary currents in the discontinuous mode. Primary current is a triangle starting from zero and rising to a level I_{p1} (point *B*) at the end of the power transistor "on" time.

At the instant of Q1 turn "off," the current I_{p1} established in the primary winding is transferred to the secondary so as to maintain the ampere-turns ratio. This current is dumped into the secondary capacitors and load during the "off" period. The secondary current ramps downward at a rate $dI_s/dt = (V_o + 1)/L_s$, where L_s is the secondary

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

inductance, which is $(N_s/N_p)^2$ times the primary magnetizing inductance. This current reaches zero at time I, leaving a dead time T_{dt} before the start of the next turn "on" period at point F. All the current and hence energy stored in the primary during the previous "on" period has now been completely delivered to the load before the next turn "on." The average DC output current will be the average of the triangle *GHI* multiplied by its duty cycle of T_{off}/T.

Now, to remain in the discontinuous mode, there must be a dead time T_{dt} (Figure 4.2*b*) between the time the secondary current has dropped to zero and the start of the next power transistor "on" time. As more power is demanded (by decreasing R_o), T_{on} must increase to keep output voltage constant (see Eq. 4.3). As T_{on} increases (at constant V_{dc}), primary current slope remains constant and the peak current rises from B to D as shown in Figure 4.2*a*. Secondary peak current ($= I_p N_p/N_s$) increases from H to K in Figure 4.2*b* and starts later in time (from G to J).

Since the output voltage is kept constant by the feedback loop, the secondary slope V_o/L_s remains constant and the point at which the secondary current falls to zero moves closer to the start of the next turn "on." This reduces T_{dt} until a point L is reached where the secondary current has just fallen to zero at the instant of the next turn "on." This load current marks the end of the discontinuous mode. Notice that if the supply voltage falls, the "on" time T_{on} must increase as V_{dc} decreases to maintain constant output voltage and this will have the same effect.

Notice that as long as the circuit is in the discontinuous mode so that a dead time always remains, increasing the "on" time increases the area of the primary and secondary current triangle *GHI* up to the limit of the area *JKL*. Further, since the DC output current is the average of the secondary current triangle multiplied by its duty cycle, then during the very next "off" period following an increase in "on" time, more secondary current is immediately available to the load.

When the dead time has been lost, however, any further increase in load current demand will increase the "on" time and decrease the "off" time as the back end of the secondary current can no longer move to the right. The secondary current will start later than point J (Figure 4.2*b*) and from a higher point than K. Then at the start of the next "on" period (position F in Figure 4.2*a* or L in Figure 4.2*b*), there is still some current or energy left in the transformer.

Now the front end of the primary current will have a small step. The feedback loop tries to deliver the increased DC load current demand by keeping the "on" time later than point J. Now at each successive "off" time, the current remaining at the end of the "off" time and hence the current step at the start of the next "on" time increase.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

Finally after many switching cycles, the front-end step of primary current and the back-end current at the end of the "off" time in Figure 4.2*d* are sufficiently high so that the area *XYZW* is somewhat larger than that sufficient to supply the output load current. Now the feedback loop starts to decrease the "on" time so that the primary trapezoid lasts from *M* to *P* and the secondary current trapezoid lasts from *T* to *W* (Figures 4.2*a* and 4.2*b*).

At this point, the volt-seconds across the transformer primary when the power transistor is "on" is equal to the "off" volt-seconds across it when the transistor is "off." For this condition, the transformer core is always reset to its original point on the hysteresis loop at the end of a full cycle. It is also the condition where the average or DC voltage across the primary is zero. This is an essential requirement, since the DC resistance in the primary is near zero and it is not possible to support a long-term DC voltage across zero resistance.

Once the continuous mode has been established, increased load current is supplied initially by an increase in "on" time (from *MP* to *MS* in Figure 4.2*c*). For fixed-frequency operation this results in a decrease in "off" time from *TW* to *XW* (Figure 4.2*d*) as the back end of the secondary current pulse cannot move further to the right in time because the dead time has vanished. Although the peak of the secondary current has increased somewhat (from point *U* to *Y*), the area lost in the decreased "off" time (*T* to *X*) is greater than the area gained in the slope change from *UV* to *YZ* in Figure 4.2*d*.

Thus, in the continuous mode, a sudden increase in DC output current initially causes a decrease in width and a smaller increase in height of the secondary current trapezoid. After many switching cycles, the average trapezoid height builds up and the width relaxes back to the point where the "on" volt-seconds again equals the "off" volt-seconds across the primary.

In addition, since the DC output voltage is proportional to the area of the secondary current trapezoid, the feedback loop, in attempting to keep the output voltage constant against an increased current demand, first drastically decreases the output voltage and then, after many switching cycles, corrects it by building up the amplitude of the secondary current trapezoid. This is the physical-circuits significance of the so-called right-half-plane-zero, which forces the drastic reduction in error-amplifier bandwidth to stabilize the feedback loop. The right-half-plane-zero will be discussed further in the chapter on loop stabilization.

After Pressman *In a fixed-frequency system, the immediate effect of increasing the "on" period (to increase primary and hence output current) will be to decrease the "off" period (the period for transfer of current to the output). Since the inductance of the transformer prevents rapid changes*

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

in current, the immediate effect of trying to increase current is to cause a short-term decrease in output current. (This is a transitory 180° phase shift between cause and effect). This short transitory phase shift is the cause of the right-half-plane-zero in the transfer function. It is a non-compensatable dynamic effect and forces the designer to provide a very low-frequency roll off in the control loop to maintain stability. Hence transient performance will not be good. The flyback converter in the continuous mode has a boost-like converter characteristic and any converter or combination of converters that have a boost-type characteristic will have the right-half-plane-zero problem. ∼K.B.

4.4.3 Continuous-Mode Flyback—Basic Operation

The flyback topology is widely used for high output voltages at relatively low power (≤5000 V at <15 W). It can also be used at powers of up to 150 W if DC supply voltages are high enough (≥160 V) so that primary currents are not excessive. The feature which makes it valuable for high output voltages is that it requires no output inductor. In forward converters, discussed above, output inductors become a troublesome problem at high output voltages because of the large voltages they have to sustain. Not requiring a high voltage free-wheeling diode is also a plus for the flyback in high voltage supplies.

After Pressman *A further advantage for high voltage applications is that relatively large voltages can be obtained with relatively fewer transformer turns. ∼K.B.*

The flyback topology is attractive for multiple output supplies because the output voltages track one another for line and load changes, far better than they do in the forward-type converters described earlier. The absence of output inductors results in better tracking. As a result, flybacks are a frequent choice for supplies with many output voltages (up to 10 isolated outputs are not uncommon). The power can be in the range of 50 to 150 W.

Although they can be used from DC input voltages as low as 5 V, it is more usual to find them used for the usual rectified 160 VDC obtained from a 115-V AC power line input. By careful design of the turns ratios, they can also be used in universal line input applications ranging from the rectified output of 160 volts DC from 110 AC inputs up to the 320 V DC obtained from a 220-V AC power line, without the need for the voltage doubling–full-wave rectifying scheme (switch S1) shown in Figure 3.1.

The latter scheme, although very widely used, has the objectionable feature that to do the switching from 115 to 220 V AC, both ends of

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

the switch in Figure 3.1 have to be accessible on the outside of the supply, which is a safety hazard. Or the supply must be opened to change the switch position. Both these alternatives have drawbacks. An alternative scheme not requiring switching will be discussed in Section 4.3.5.

Both modes have an identical circuit diagram, shown in Figure 4.1, and it is only the transformer's magnetizing inductance and output load current that determines the operating mode. It has been shown that with a given magnetizing inductance, a circuit that has been designed for the discontinuous mode will move into the continuous mode when the output load current is increased beyond a unique boundary. The mechanism for this and its consequence are discussed in more detail below.

The discontinuous mode (as shown in Figure 4.2*a*) does not have a front-end step in the primary current. At turn "off" (as shown in Figure 4.2*b*), the secondary current will be a decaying triangle that has ramped down to zero before the next turn "on." All the energy stored in the primary during the "on" period has been completely delivered to the secondary and thus to the load before the next turn "on."

In the continuous mode, however (as seen in Figure 4.2*c*), the primary current does have a front-end step and the characteristic of a rising current ramp following the step. During the "off" period of *Q*1 (Figure 4.2*d*), the secondary current has the shape of a decaying triangle sitting on a step with current still remaining in the secondary at the instant of the next turn "on" action. Clearly there is still some energy left in the transformer at the instant of the next turn "on."

The two modes have significantly different operating properties and usages. The discontinuous mode, which does not have a right-half-plane-zero in the transfer function, responds more rapidly to transient load changes with a lower transient output voltage spike.

A penalty is paid for this performance, in that the secondary peak current in the discontinuous mode can be between two and three times greater than that in the continuous mode. This is shown in Figures 4.2*b* and 4.2*d*. Secondary DC load current is the average of the current waveshapes in those figures. Also, assuming closely equal "off" times, it is obvious that the triangle in the discontinuous mode must have a much larger peak than the trapezoid of the continuous mode for the two waveshapes to have equal average values.

With larger peak secondary currents, the discontinuous mode has a larger transient output voltage spike at the instant of turn "off" (Section 4.3.4.1) and requires a larger *LC* spike filter to remove it.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Also, the larger secondary peak current at the start of turn "off" in the discontinuous mode causes a greater RFI problem. Even for moderate output powers, the very large initial spike of secondary current at the instant of turn "off" causes a much more severe noise spike on the output ground bus, because of the large *di/dt* into the output bus inductance.

After Pressman *A major advantage of the discontinuous mode is that the secondary rectifier diodes turn "off" under low current stress conditions. Also they are fully "off" before the next "on" edge of* Q1. *Hence the problem of diode reverse recovery is eliminated. This is a major advantage in high voltage applications as diode reverse recovery current spikes are difficult to eliminate and are a rich source of RFI. ~K.B.*

Due to the poor form factor, secondary RMS currents in the discontinuous mode can be much larger than those in the continuous mode. Hence, the discontinuous mode requires larger secondary wire size and output filter capacitors with larger ripple current ratings. The rectifier diodes will also run hotter in the discontinuous mode because of the larger secondary RMS currents. Further, the primary peak currents in the discontinuous mode are larger than those in the continuous mode. For the same output power, the triangle of Figure 4.2*a* must have a larger peak than the trapezoid of Figure 4.2*c*. The consequence is that the discontinuous mode with its larger peak primary current requires a power transistor of higher current rating and possibly higher cost. Also, the higher primary current at the turn "off" edge of *Q*1 results in a potential for greater RFI problems.

Despite all the disadvantages of the discontinuous mode, it is much more widely used than the continuous mode. This is so for two reasons. First, as mentioned above, the discontinuous mode, with an inherently smaller transformer magnetizing inductance, responds more quickly and with a lower transient output voltage spike to rapid changes in output load current or input voltage. Second, because of a unique characteristic of the continuous mode (its transfer function has a right-half-plane-zero, to be discussed in a later chapter on feedback loop stabilization), the error amplifier bandwidth must be drastically reduced to stabilize the feedback loop.

After Pressman *Modern power devices, such as the Power Integration's Top Switch range of products, have the "noisy" FET drain part of the chip isolated from the heat sink tab. This, together with integrated drive and control circuits, which further reduce radiating area, very much reduces the RFI problems normally associated with the discontinuous flyback topology. ~K.B.*

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

4.5 Design Relations and Sequential Design Steps

4.5.1 Step 1: Establish the Primary/Secondary Turns Ratio

For the most expedient design, there are a number of decisions that should be made in the following logical sequence.

First select a core size to meet the power requirements.

Next choose the primary/master secondary turns ratio N_p/N_{sm} to determine the maximum "off"-voltage stress $\overline{V_{\mathrm{ms}}}$ on the power transistor in the absence of a leakage inductance spike as follows:

Neglecting the leakage spike, the maximum transistor voltage stress at maximum DC input $\overline{V_{\mathrm{dc}}}$ and for a 1-V rectifier drop is

$$\overline{V_{\mathrm{ms}}} = \overline{V_{\mathrm{dc}}} + \frac{N_p}{N_{\mathrm{sm}}}(V_o + 1) \tag{4.4}$$

where $\overline{V_{\mathrm{ms}}}$ is chosen sufficiently low so that a leakage inductance spike of $0.3V_{\mathrm{dc}}$ on top of that still leaves a safety margin of about 30% below the maximum pertinent transistor rating (V_{ceo}, V_{cer}, or V_{cev}).

4.5.2 Step 2: Ensure the Core Does Not Saturate and the Mode Remains Discontinuous

To ensure that the core does not drift up or down its hysteresis loop, the "on" volt-second product ($A1$ in Figure 4.1d) must equal the reset volt-second product ($A2$ in Figure 4.1d). Assume that the "on" drop of $Q1$ and the forward drop of the rectifier $D2$ are both 1 V:

$$(\underline{V_{\mathrm{dc}}} - 1)\overline{T_{\mathrm{on}}} = (V_o + 1)\frac{N_p}{N_{\mathrm{sm}}}T_r \tag{4.5}$$

where T_r shown in Figure 4.1c is the reset time required for the secondary current to return to zero.

To ensure the circuit operates in the discontinuous mode, a dead time (T_{dt} in Figure 4.1c) is established so that the maximum "on" time $\overline{T_{\mathrm{on}}}$, which occurs when V_{dc} is a minimum, plus the reset time T_r is only 80% of a full period. This leaves $0.2T$ margin against unexpected decreases in R_o, which according to Eq. 4.3 would force the feedback loop to increase T_{on} in order to keep V_o constant.

As for the boost regulator, which is also a flyback type (Sections 1.4.2 and 1.4.3), it was pointed out that if the error amplifier has been designed to keep the loop stable only in the discontinuous mode, it may break into oscillation if the circuit momentarily enters the continuous mode.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

Increasing DC load current or decreasing V_{dc} causes the error amplifier to increase T_{on} in order to keep V_o constant (Eq. 4.3). This increased T_{on} eats into the dead time T_{dt}, and eventually the secondary current does not fall to zero by the start of the next $Q1$ "on" time. This is the start of the continuous mode, and if the error amplifier has not been designed with a drastically lower bandwidth than required for discontinuous mode, the circuit will oscillate. To ensure that the circuit remains discontinuous, the maximum "on" time that will generate the desired maximum output power is established:

$$\overline{T_{on}} + T_r + T_{dt} = T$$

or

$$\overline{T_{on}} + T_r = 0.8T \tag{4.6}$$

Now in Eqs. 4.5 and 4.6, there are two unknowns, as N_p/N_{sm} has been calculated from Eq. 4.4 for specified $\overline{V_{dc}}$ and $\overline{V_{ms}}$. Then from the last two relations

$$\overline{T_{on}} = \frac{(V_o + 1)(N_p/N_{sm})(0.8T)}{(\underline{V_{dc}} - 1) + (V_o + 1)(N_p/N_{sm})} \tag{4.7}$$

4.5.3 Step 3: Adjust the Primary Inductance Versus Minimum Output Resistance and DC Input Voltage

From Eq. 4.3, the primary inductance is

$$L_p = \frac{R_o}{2.5T}\left(\frac{\underline{V_{dc}}\,\overline{T_{on}}}{V_o}\right)^2 = \frac{(\underline{V_{dc}}\,\overline{T_{on}})^2}{2.5T\,\overline{P_o}} \tag{4.8}$$

4.5.4 Step 4: Check Transistor Peak Current and Maximum Voltage Stress

If the transistor is a bipolar type, it must have an acceptably high gain at the peak current operating current Ip. This is

$$I_p = \frac{\underline{V_{dc}}\,\overline{T_{on}}}{L_p} \tag{4.9}$$

where $\underline{V_{dc}}$ is specified and $\overline{T_{on}}$ is calculated from Eq. 4.7 and L_p is calculated from Eq. 4.8.

If $Q1$ is a MOSFET, it should have a peak current rating about 5 to 10 times the value calculated from Eq. 4.9 so that its "on"-state resistance is low enough to yield an acceptably low voltage drop and power loss.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

4.5.5 Step 5: Check Primary RMS Current and Establish Wire Size

The primary current is a triangle of peak amplitude I_p (Eq. 4.9) at a maximum duration $\overline{T_{\text{on}}}$ out of every period T. Its RMS value (Section 2.2.10.6) is

$$I_{\text{rms(primary)}} = \frac{I_p}{\sqrt{3}}\sqrt{\frac{\overline{T_{\text{on}}}}{T}} \tag{4.10}$$

where I_p and $\overline{T_{\text{on}}}$ are as given by Eqs. 4.9 and 4.7.

At 500 circular mils per RMS ampere, the required number of circular mils is

$$\begin{aligned}\text{Circular mils required (primary)} &= 500\, I_{\text{rms(primary)}} \\ &= 500\frac{I_p}{\sqrt{3}}\sqrt{\frac{\overline{T_{\text{on}}}}{T}}\end{aligned} \tag{4.11}$$

4.5.6 Step 6: Check Secondary RMS Current and Select Wire Size

The secondary current is a triangle of peak amplitude $I_s = I_p(N_p/N_s)$ and duration T_r. Primary/secondary turns ratio N_p/N_s is given by Eq. 4.4 and $T_r = (T - T_{\text{on}})$. Secondary RMS current is then

$$I_{\text{rms(secondary)}} = \frac{I_p(N_p/N_s)}{\sqrt{3}}\sqrt{\frac{T_r}{T}} \tag{4.12}$$

At 500 circular mils per RMS ampere, the required number of circular mils is

$$\text{Secondary circular mils required} = 500 I_{\text{rms(secondary)}} \tag{4.13}$$

4.6 Design Example for a Discontinuous-Mode Flyback Converter

We will now look at a worked design example for a flyback converter with the following specifications:

V_o	5.0 V
$P_{o(\text{max})}$	50 W
$I_{o(\text{max})}$	10 A
$I_{o(\text{min})}$	1.0 A
$V_{\text{dc(max)}}$	60 V
$V_{\text{dc(min)}}$	38 V
Switching frequency	50 kHz

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

First, select the voltage rating of the transistor as this mainly determines the transformer turns ratio. Choose a device with a 200-V rating. In Eq. 4.4 choose the maximum stress V_{ms} on the transistor in the "off" state (excluding the leakage inductance spike) as 120 V. Then even with a 25% or 30-V leakage spike, this leaves a 50-V margin to the maximum voltage rating. Then from Eq. 4.4

$$120 = 60 + \frac{N_p}{N_{sm}}(V_o + 1) \quad \text{or} \quad \frac{N_p}{N_{sm}} = 10$$

Now choose maximum "on" time from Eq. 4.7:

$$\begin{aligned}\overline{T_{om}} &= \frac{(V_o + 1)(N_p/N_{sm})(0.8T)}{(V_{dc} - 1) + (V_o + 1)N_p/N_{sm}} \\ &= \frac{6 \times 10 \times 0.8 \times 20}{(38 - 1) + 6 \times 10} \\ &= 9.9\,\mu\text{s}\end{aligned}$$

From Eq. 4.8

$$\begin{aligned}L_p &= \frac{(V_{dc}\overline{T_{on}})^2}{2.5T P_o} \\ &= \frac{(38 \times 9.9 \times 10^{-6})^2}{2.5 \times 20 \times 10^{-6} \times 50} \\ &= 56.6\,\mu\text{H}\end{aligned}$$

From Eq. 4.9

$$\begin{aligned}I_p &= \frac{V_{dc}\overline{T_{on}}}{L_p} \\ &= \frac{38 \times 9.9 \times 10^{-6}}{56.6 \times 10^{-6}} \\ &= 6.6 \text{ A}\end{aligned}$$

From Eq. 4.10, primary RMS current is

$$\begin{aligned}I_{rms(primary)} &= \frac{I_p}{\sqrt{3}}\sqrt{\frac{\overline{T_{on}}}{T}} \\ &= \frac{6.6}{\sqrt{3}} \times \sqrt{\frac{9.9}{20}} \\ &= 2.7 \text{ A}\end{aligned}$$

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

From Eq. 4.11, primary circular mils requirement is

$$I_{\text{(primary circular mils)}} = 500 \times 2.7 = 1350 \text{ circular mils}$$

This calls for a No. 19 wire of 1290 circular mils, which is close enough.

From Eq. 4.12, secondary RMS current is

$$I_{\text{rms(secondary)}} = \frac{I_p(N_p/N_s)}{\sqrt{3}}\sqrt{\frac{T_r}{T}}$$

But reset time T_r is

$$(0.8T - \overline{T_{\text{on}}}) = (16 - 9.9) = 6.1\,\mu\text{s}$$

Then

$$I_{\text{rms(secondary)}} = \frac{6.6 \times 10}{\sqrt{3}}\sqrt{\frac{6.1}{20}}$$
$$= 21 \text{ A}$$

We see that from Eq. 4.12, the required number of circular mils is $500 \times 21 = 10{,}500$. This calls for No. 10 wire, which is impractically large in diameter. A foil winding or a number of smaller diameter wires in parallel with an equal total circular-mil area would be used.

After Pressman *Contrary to popular belief, the wire size and leakage inductance in a flyback transformer are important design parameters. Multiple strands of wire in parallel are required, with the maximum wire size selected to minimize skin and proximity effects. Even though the secondary energy comes from the energy stored in the transformer core, leakage inductance must still be minimized to ensure good energy transfer from the primary winding to the secondary windings to reduce voltage spikes on* Q1 *at the "off" transition. This will reduce the amount of snubbing required on* Q1 *and reduce RFI. (See Chapter 7.) ~K.B.*

The output capacitor is chosen on the basis of specified peak-to-peak output voltage ripple as follows:

At maximum output current, the filter capacitor C_o carries the 10-A output current for all but the 6.1 μs reset period, or 13.9 μs. The voltage on this capacitor droops by $V = I(T - t_{\text{off}})/C_o$. Then for a voltage droop of 0.05 V

$$C_o = \frac{10 \times 13.9 \times 10^{-6}}{0.05}$$
$$= 2800\,\mu\text{F}$$

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

From Section 1.4.7, the average ESR of a 2800-μF aluminum electrolytic capacitor is

$$R_{\text{esr}} = 65 \times 10^{-6}/C_o = 0.023\ \Omega$$

At the instant of transistor turn "off," a peak secondary current of 66 amps flows through the above capacitor ESR, causing a thin spike of 66 × 0.023 = 1.5 V. This large-amplitude thin spike at transistor turn "off" is a universal problem with flybacks having a large N_p/N_s ratio. It is usually solved by using a larger filter capacitor than calculated as above (since R_{esr} is inversely proportional to C_o) and/or integrating away the thin spike with a small *LC* circuit.

After Pressman *Since the spike contains a large amount of high frequency components, combining several capacitors in parallel will reduce the spike amplitude significantly. Small ceramic and film caps are often used. ~K.B.*

Selecting a transformer core for a flyback topology circuit is significantly different than selecting for a forward converter. Remember in the flyback that when current flows in the primary, the secondary current is zero, and there is no current flow in the secondary to buck out the primary ampere turns as there is in the forward converters. Thus in the flyback, all the primary ampere turns tend to saturate the core.

In contrast, in non-flyback topologies, secondary load current flows when primary current flows and is in the direction (by Lenz's law) to cancel the ampere turns of the primary. It is only the primary magnetizing current that drives the core over its hysteresis loop and moves it toward saturation. That magnetizing current is kept a small fraction of the primary load current by providing a large magnetizing inductance and hence core saturation is not a basic a problem with non-flyback topologies.

Hence, flyback transformer cores must have some means of carrying large primary currents without saturating. This is done by choosing low permeability materials such as MPP (molybdenum permalloy powder) cores that have an inherent air gap or by using gapped ferrite cores (Section 2.3.9.3 and Chapter 7). This is discussed further in the following section.

4.6.1 Flyback Magnetics

Referring to Figure 4.1*a*, it is seen from the winding dots that when the transistor is "on" and current flows in the primary, no secondary currents flow. This is totally different from forward-type converters,

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

in which current flows in the secondary when it flows in the primary. Thus in a forward-type converter, primary current flows into a dot end and secondary current flows out of a dot end.

Primary and secondary load ampere-turns then cancel each other out and do not move the core across its hysteresis loop. In the forward-type converters, it is only the magnetizing current that drives the core across the hysteresis loop and may potentially saturate it. But this magnetizing current is a small fraction (rarely >10%) of the total primary current.

In a flyback converter, however, the entire triangle of primary current shown in Figure 4.1*b* drives the core across the hysteresis loop as it is not canceled out by any secondary ampere turns. Thus, even at very low output power, an ungapped ferrite core would almost immediately saturate and destroy the transistor if nothing were done to prevent it.

To prevent core saturation in the flyback transformer, the core is gapped. The gapped core can be either of two types. It can be a solid ferrite core with a known air-gap length obtained by grinding down the center leg in EE or cup-type cores. The known gap length can also be obtained by inserting plastic shims between the two halves of an EE, cup, or UU core.

A more usual gapped core for flyback converters is the MPP or molypermalloy powder core. Such cores are made of a baked and hardened mix of magnetic powdered particles. These powdered particles are mixed in a slurry with a plastic resin binder and cast in the shape of a toroid. Each magnetic particle in the toroid is thus encapsulated within a resin envelope that behaves as a "distributed air gap" and acts to keep the core from saturating. The basic magnetic material that is ground up into a powder is Square Permalloy 80, an alloy of 79% nickel, 17% iron, and 4% molybdenum, made by Magnetics Inc. and Arnold Magnetics, among others.

The permeability of the resulting toroid is determined by controlling the concentration of magnetic particles in the slurry. Permeabilities are controlled to within ±5% over large temperature ranges and are available in discrete steps ranging from 14 to 550. Toroids with low permeability behave like gapped cores with large air gaps. They require a relatively large number of turns to yield a desired inductance but tolerate many ampere-turns before they saturate. Higher permeability cores require relatively fewer turns but saturate at a lower number of ampere-turns.

Such MPP cores are used not only for flyback transformers in which all the primary current is DC bias current. They are also used for forward converter output inductors where, as has been seen, a unique inductance is required at the large DC output current bias (Section 1.3.6).

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

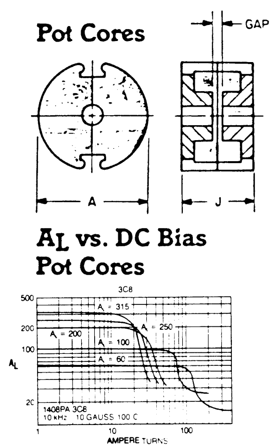
4.6.2 Gapping Ferrite Cores to Avoid Saturation

Adding an air gap to a solid ferrite core achieves two results. First, it tilts the hysteresis loop as shown in Figure 2.5 and hence decreases its permeability, which must be known to select the number of turns for a desired inductance. Second, and more important, it increases the number of ampere turns it can tolerate before it saturates.

Core manufacturers often offer curves that permit calculation of the number of turns for a desired inductance and the number of ampere-turns at which saturation commences. Such curves are shown in Figure 4.3 and show A_{lg}, the inductance per 1000 turns with an air

Part No	A	J	Approx Gap	A_L mH/ 1000 Turns ±3%	μ_e (Ref.)
1408PA 60-3C8	.551	.328	.027	60	37
1408PA100-3C8	.551	.328	.013	100	63
1408PA200-3C8	.551	.328	.005	200	126
1408PA250-3C8	.551	.328	.004	250	156
1408PA315-3C8	.551	.328	.003	315	198
1811PA 75-3C8	.705	.416	.037	75	35
1811PA130-3C8	.705	.416	.018	130	62
1811PA250-3C8	.705	.416	.008	250	119
1811PA315-3C8	.705	.416	.006	315	150
1811PA400-3C8	.705	.416	.004	400	190
2213PA 85-3C8	.846	.528	.050	85	33
2213PA145-3C8	.846	.528	.025	145	57
2213PA315-3C8	.846	.528	.009	315	123
2213PA400-3C8	.846	.528	.007	400	157
2213PA500-3C8	.846	.528	.005	500	196
2616PA100-3C8	1.004	.634	.064	100	31
2616PA170-3C8	1.004	.634	.032	170	53
2616PA400-3C8	1.004	.634	.009	400	125
2616PA500-3C8	1.004	.634	.007	500	156
2616PA630-3C8	1.004	.634	.005	630	197
3019PA125-3C8	1.181	.740	.070	125	32
3019PA210-3C8	1.181	.740	.035	210	54
3019PA500-3C8	1.181	.740	.011	500	129
3019PA630-3C8	1.181	.740	.008	630	163
3019PA800-3C8	1.181	.740	.007	800	206
3622PA160-3C8	1.398	.855	.079	160	33
3622PA275-3C8	1.398	.855	.040	275	57
3622PA630-3C8	1.398	.855	.016	630	131
3622PA800-3C8	1.398	.855	.012	800	166
3622PA1000-3C8	1.398	.855	.008	1000	208
4229PA160-3C8	1.669	1.164	.103	160	33
4229PA275-3C8	1.669	1.164	.051	275	56
4229PA630-3C8	1.669	1.164	.020	630	128
4229PA800-3C8	1.669	1.164	.015	800	162
4229PA1000-3C8	1.669	1.164	.011	1000	202

Nominal dimensions in inches.



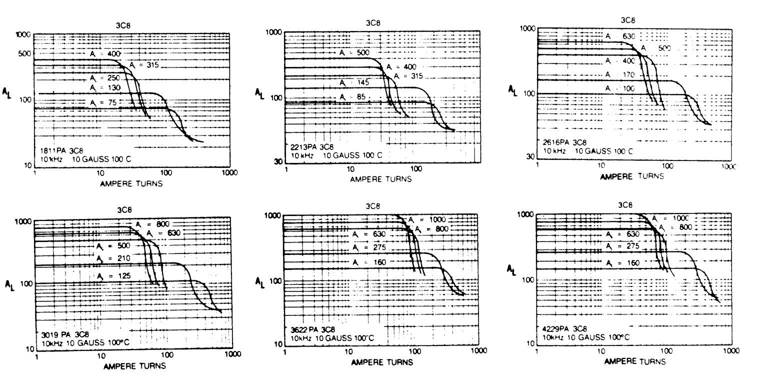


FIGURE 4.3 Inductance per 1000 turns (A_{lg}) for various ferrite cores with various air gaps. Note the "cliff" points in ampere-turns where saturation commences. (*Courtesy Ferroxcube Corporation.*)

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

gap and the number of ampere-turns (NI_{sat}) where saturation starts to set in. Since inductance is proportional to the square of the number of turns, the number of turns N_l for any inductance L is calculated from

$$N_l = 1000\sqrt{\frac{L}{A_{\text{lg}}}} \tag{4.14}$$

Figure 4.3 shows A_{lg} curves for a number of different air gaps and the "cliff" point at which saturation starts. It can be seen that the larger the air gap, the lower the value of A_{lg} and the larger the number of ampere-turns at which saturation starts. If such curves were available for all cores at various air gaps, Eq. 4.14 would give the number of turns for any selected air gap from the value of A_{lg} read from the curve. The cliff point on the curve would tell whether, at those turns and for the specified primary current, the core had fallen over the saturation cliff.

Such curves, though, are not available for all cores and all air gaps. This is no problem, because A_{lg} can be calculated with reasonable accuracy from Eq. 2.39 using A_l with no gap, which is always given in the manufacturers' catalogs. The cliff point at which saturation starts can be calculated from Eq. 2.37 for any air gap. The cliff point corresponds to the flux density in iron B_i, where the core material itself starts bending over into saturation.

From Figure 2.3, it is seen that this is not a very sharp breaking point, but occurs around 2500 G for this ferrite material (Ferroxcube 3C8). Thus the cliff in ampere-turns is found by substituting 2500 G in Eq. 2.37. As noted in connection with Eq. 2.37, in the usual case, the air-gap length l_a is much larger than l_i/u as u is so large. Then the iron flux density as given by Eq. 2.37 is determined mainly by the air-gap length l_a.

4.6.3 Using Powdered Permalloy (MPP) Cores to Avoid Saturation

These toroidal cores are widely used and made by Magnetics Inc. (data in catalog MPP303S) and by Arnold Co. (data in catalog PC104G).

After Pressman *The term* transformer *in the phrase* flyback transformer *is a misnomer and is very misleading. For true transformer action to take place both primary and secondaries must conduct current at the same time. We are all aware that a true transformer conserves the primary to secondary voltage ratio (irrespective of current). In the flyback case the so-called transformer conserves the primary to secondary ampere-turns ratio (irrespective of voltage). This means it is really a "choke"—an inductor with a DC component of current and additional windings. I find it much easier and less confusing to design my flyback "transformers" from this perspective. The reader may also find it helpful to use this approach because the inductance*

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

becomes the dependant variable and can be easily adjusted to get the desired results. Chapter 7 deals with the design in this way.

You will see from the following that although he does not mention it, Pressman is leading you in the direction of choke design. ∼K.B.

The problem in designing a core of desired inductance at a specified maximum DC current bias is to select a core geometry and material permeability, such that the core does not saturate at the maximum ampere-turns to which it is subjected. There are a limited number of core geometries, each available in permeabilities ranging from 14 to 550. Selection procedures are described in the catalogs mentioned above, but the following has been found more direct and useful.

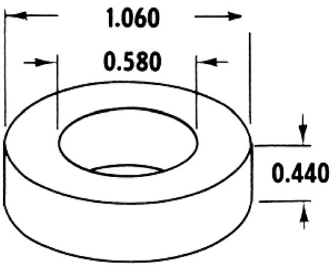
In the Magnetics Inc. catalog, one full page (Figure 4.4) is devoted to each size toroid, and for each size, its A_l value (inductance in millihenries per 1000 turns) is given for each discrete permeability. Figure 4.5, also from the Magnetics Inc. catalog, gives the falloff in permeability (or A_l value) for increasing magnetizing force in oersteds for core materials of the various available permeabilities. (Recall the oersted–ampere-turns relation in Eq. 2.6.)

A core geometry and permeability can be selected so that at the maximum DC current and the selected number of turns, the A_l and hence inductance has fallen off by any desired percentage given in Figure 4.5. Then at zero DC current, the inductance will be greater by that percentage. Such inductors or chokes are referred to as "swinging chokes" and in many applications are desirable. For example, if an inductor is permitted to swing a great deal, in an output filter, it can tolerate a very low minimum DC current before it goes discontinuous (Section 1.3.6). But this greatly complicates the feedback-loop stability design and, most often, the inductor in an output filter or transformer in a flyback will not be permitted to "swing" or vary very much between its zero and maximum current value.

Referring to Figure 4.4, it is seen that a core of this specific size is available in permeabilities ranging from 14 to 550. Cores with permeability above 125 have large values of A_l and hence require fewer turns for a specified inductance at zero DC current bias. But in Figure 4.5 it is seen that the higher-permeability cores saturate at increasingly lower ampere-turns of bias. Hence in power supply usage, where DC current biases are rarely under 1 A, cores of permeability greater than 125 are rarely used, and an inductance swing or change of 10% from zero to the maximum specified current is most often acceptable.

In Figure 4.5, it is seen that for a permeability dropoff or swing of 10%, core materials of permeabilities 14, 26, 60, and 125 can sustain maximum magnetizing forces of only 170, 95, 39, and 19 Oe, respectively. These maximum magnetizing forces in oersteds can be translated into maximum ampere-turns by Eq. 2.6 ($\overline{H} = 0.4\pi(\overline{NI})/l_m$), in

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.



WINDOW AREA	308,000 cir. mils	
CROSS SECTION	0.1014 in²	0.654 cm²
PATH LENGTH	2.50 in	6.35 cm
WEIGHT	1.3 oz (.08 lb.)	36. gm

WINDING TURN LENGTH

WINDING FACTOR	LENGTH/TURN	
100% (UNITY)	0.1714 ft	5.23 cm
60%	0.1526 ft	4.66 cm
40%	0.1344 ft	4.10 cm
20%	0.1263 ft	3.85 cm
0%	0.1233 ft	3.76 cm

CORE DIMENSIONS AFTER FINISH

OD (MAX.)	1.090 in	27.7 mm
ID (MIN.)	0.555 in	14.10 mm
HT (MAX.)	0.472 in	11.99 mm

WOUND COIL DIMENSIONS

UNITY WINDING FACTOR

OD (MAX.)	1.468 in	37.3 mm
HT (MAX.)	0.944 in	24.0 mm

MAGNETIC INFORMATION

PART NO.	PERM μ	INDUCTANCE @ 1000 TURNS MH±8%	NOMINAL DC RESISTANCE OHMS/MH**	FINISHES AND STABILIZATIONS*	GRADING STATUS 2% BANDS	B/NI GAUSS PER AMP. TURN
55933–	14	18	0.457	A2	*	2.77 (<1500 gauss)
55932–	26	32	0.257	A2	*	5.15 (<1500 gauss)
55894–	60	75	0.110	ALL	YES	11.9 (<1500 gauss)
55930–	125	157	0.0524	ALL	YES	24.8 (<1500 gauss)
55929–	147	185	0.0444	ALL	YES	29.1 (<1500 gauss)
55928–	160	201	0.0409	ALL	YES	31.7 (<1500 gauss)
55924–	173	217	0.0379	ALL	YES	34.3 (<1500 gauss)
55927–	200	251	0.0327	ALL	YES	39.6 (<600 gauss)
55925–	300	377	0.0218	A2 and L6	YES	59.4 (<300 gauss)
55926–	550	740	0.0111	A2	YES	109 (<50 gauss)

14, 26, 60 and 125μ types are available as high flux cores. See p. 84, "How to Order".

WINDING INFORMATION

FOR UNITY WINDING FACTOR

AWG WIRE SIZE	TURNS	Rdc OHMS	AWG WIRE SIZE	TURNS	Rdc OHMS
9	21	0.00291	24	587	2.58
10	27	0.00459	25	725	4.02
11	34	0.00726	26	906	6.37
12	42	0.01148	27	1,141	10.05
13	53	0.01805	28	1,400	15.67
14	66	0.0284	29	1,711	24.1
15	82	0.0447	30	2,139	38.1
16	103	0.0707	31	2,633	59.1
17	127	0.1102	32	3,209	89.1
18	159	0.1739	33	3,980	140.5
19	197	0.272	34	5,066	227
20	246	0.428	35	6,286	357
21	308	0.676	36	7,759	552
22	380	1.056	37	9,478	832
23	474	1.649	38	11,847	1,316

FIGURE 4.4 A typical MPP core. With its large distributed air gap, it can tolerate a large DC current bias without saturating. It is available in a large range of different geometries. (*Courtesy Magnetics Inc.*)

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

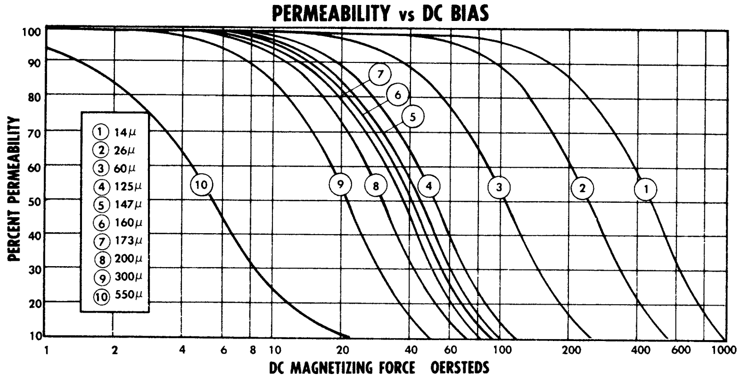


FIGURE 4.5 Falloff in permeability of A_1 for MPP cores of various permeabilities versus DC magnetizing force in oersteds. (*Courtesy Magnetics Inc.*)

which l_m is the magnetic path length in centimeters, given in Figure 4.3 for this particular core geometry as 6.35 cm.

From these maximum numbers of ampere-turns ($\overline{NI}$), beyond which inductance falls off more than 10%, the maximum number of turns ($\overline{N}$) is calculated for any peak current. From $\overline{N}$, the maximum inductance possible for any core at the specified peak current is calculated as $L_{\max} = 0.9A_1(N_{\max}/1000)^2$.

Tables 4.1, 4.2, and 4.3 show $N_{\max}$ and $L_{\max}$ for three often-used core geometries in permeabilities of 14, 26, 60, and 125 at peak currents of 1, 2, 3, 5, 10, 20, and 50 amperes. These tables permit core geometry and permeability selection at a glance without iterative calculations.

Table 4.1 is used in the following manner. Assume that this particular core has the acceptable geometry. The table is entered horizontally to the first peak current greater than specified value. At that peak current, move down vertically until the first inductance $L_{\max}$ greater than the desired value is reached. The core at that point is the only one which can yield the desired inductance with only a 10% swing. The number of turns N_d on that core for a desired inductance L_d within 5% is given by

$$N_d = 1000\sqrt{\frac{L_d}{0.95A_l}}$$

where A_l is the value in column 3 in Table 4.1. If, moving vertically, no core can be found whose maximum inductance is greater than the desired value, the core with the next larger geometry (greater OD or greater height) must be used.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Magnetics Inc. core number	Perme-ability	A_l, mH per 1000 turns	Maximum $\overline{H}$ for 10% falloff in inductance	$\overline{NI}$ Maximum permissible ampere-turns corresponding to $\overline{H}$	Maximum permissible turns and inductance at those turns for a 10% inductance falloff at indicated peak currents							
						N_{max}/L_{max}						
Core	μ	A_l	H	NI	1A	2A	3A	5A	10A	20A	50A	I_p
55930	125	157	19	96	96	48	32	19	10	5	2	N_{max}
					1,382	339	145	56	15	3.5	0.6	L_{max}
55894	60	75	39	197	197	99	66	39	20	10	4	N_{max}
					2,620	662	294	103	27	7	1	L_{max}
55932	26	32	95	480	480	240	160	96	48	24	10	N_{max}
					6,635	1,659	737	265	66	17	3	L_{max}
55933	14	18	170	859	859	430	286	172	86	43	17	N_{max}
					11,954	2,995	1,325	479	120	30	5	L_{max}

Note: Magnetics Inc. MPP cores. All cores have outer diameter (OD) = 1.060 in, inner diameter (ID) = 0.58 in, height = 0.44 in, l_m = 6.35 cm. All inductances in microhenries.

TABLE 4.1 Maximum number of turns yielding maximum inductance for various peak currents I_p at maximum inductance falloff of 10% from zero current

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Magnetics Inc. core number	Perme-ability	A_l, mH per 1000 turns	Maximum $\overline{H}$ for 10% falloff in inductance	$\overline{NI}$ Maximum permissible ampere-turns corresponding to $\overline{H}$	Maximum permissible turns and inductance at those turns for a 10% inductance falloff at indicated peak currents							
						N_{max}/L_{max}						
Core	µ	A_l	H	NI	1A	2A	3A	5A	10A	20A	50A	I_p
55206	125	68	19	77	77	39	26	15	8	4	2	N_{max}
					363	93	41	14	4	1	0.24	L_{max}
55848	60	32	39	158	158	79	53	32	16	8	3	N_{max}
					719	180	81	29	7	2	0.26	L_{max}
55208	26	14	95	385	385	193	128	77	39	19	8	N_{max}
					1,868	469	206	75	19	4.5	0.8	L_{max}
55209	14	7.8	170	689	689	345	230	138	69	34	14	N_{max}
					3,333	836	371	134	33	8	1.4	L_{max}

Note: Magnetics Inc. MPP cores: OD = 0.8 in, ID = 0.5 in, height = 0.25 in, l_m = 5.09 cm. All inductances in microhenries.

TABLE 4.2 Maximum number of turns and maximum inductance for various peak currents I_p at a maximum inductance falloff of 10% from zero current

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Core	μ	A_l	$\overline{H}$	$\overline{NI}$	1A	2A	3A	5A	10A	20A	50A	I_p
55438	125	281	19	162	162	81	54	32	16	8	3	N_{max}
					6,637	1,659	737	259	65	16	2	L_{max}
55439	60	135	39	333	333	167	111	67	33	17	7	N_{max}
					13,473	3,389	1,497	545	132	35	6	L_{max}
55440	26	59	95	812	812	406	271	162	81	41	16	N_{max}
					35,011	8,753	3,900	1,394	348	89	14	L_{max}
55441	14	32	170	1454	1,454	727	485	291	145	73	29	N_{max}
					60,744	15,222	6,774	2,439	605	153	24	L_{max}

Note: Magnetics Inc. MPP cores: OD = 1.84 in, ID = 0.95 in, height = 0.71 in, l_m = 10.74 in. All inductances in microhenries.

TABLE 4.3 Maximum number of turns and maximum inductance for various peak currents I_p at a maximum inductance falloff of 10% from zero current

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

The core ID must be large enough to accommodate the number of turns of wire selected at the rate of 500 circular miles per RMS ampere, or the next larger size core must be used.

Tables 4.2 and 4.3 show similar data for smaller (OD = 0.80 in) and larger (OD = 1.84 in) families of cores. Similar charts can be generated for all the other available core sizes, but Tables 4.1 to 4.3 bracket about 90% of the possible designs for flyback transformers under 500 W or output inductors of up to 50 A.

A commonly used scheme for correcting the number of turns on a core when an initial selection has resulted in too large an inductance falloff should be noted. If, for an initially selected number of turns and a specified maximum current, the inductance or permeability falloff from Figure 4.5 is down by P%, the number of turns is increased by P%.

This moves the operating point further out by P%, as the magnetizing force in oersteds is proportional to the number of turns. The core slides further down its saturation curve, and it might be thought that the inductance would fall off even more. But since inductance is proportional to the square of the number of turns, and magnetizing force is proportional only to the number of turns, the zero current inductance has been increased by $2P$% and magnetizing force has gone up only by P%. The inductance is then correct at the specified maximum current. If the consequent swing is too large, a larger core must be used.

4.6.4 Flyback Disadvantages

Despite its many advantages, the flyback has the following drawbacks.

4.6.4.1 Large Output Voltage Spikes

At the end of the "on" time, the peak primary current is given by Eq. 4.9. Immediately after the end of the "on" time, that primary peak current, multiplied by the turns ratio N_p/N_s, is driven into the secondary where it decays linearly as shown in Figure 4.1*c*. In most cases, output voltages are low relative to input voltage, resulting in a large N_p/N_s ratio and a consequent large secondary current.

At the start of turn "off," the impedance looking into C_o is much lower than R_o (Figure 4.1) and almost all the large secondary current flows into C_o and its equivalent series resistor R_{esr}. This produces a large, thin output voltage spike, $I_p(N_p/N_s)R_{esr}$. The spike is generally less than 0.5 μs in width, as it is differentiated with a time constant of $R_{esr}C_o$.

Frequently a power supply specification calls for output voltage ripple only as an RMS or peak-to-peak fundamental value. Such a large, thin spike has a very low RMS value and, if a sufficiently large

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

output filter capacitor is chosen, the supply can easily meet its RMS ripple specification but can have disastrously high, thin output spikes. It is common to see a 50-mV fundamental peak-to-peak output ripple with a 1-V thin spike sitting on top of it.

Thus, a small *LC* filter is almost always added after the main storage capacitor in flybacks. The *L* and *C* can be quite small as they have to filter out a spike generally less than 0.5 μs in width. The inductor is usually considerably smaller than the inductor in forward-type converters, but it still has to be stocked, and board space must be provided for it. Output voltage sensing for the error amplifier is taken before this *LC* filter.

4.6.4.2 Large Output Filter Capacitor and High Ripple Current Requirement

A filter capacitor for a flyback must be much larger than for a forward-type converter. In a forward converter, when the power transistor turns "off" (Figure 2.10), load current is supplied from the energy stored in both the filter inductor and capacitor. But in the flyback, that capacitor is necessarily larger because it is the stored energy in it alone that supplies current to the load during the transistor "on" time. Output ripple is determined mostly by the ESR of the filter capacitor (see Section 1.3.7). An initial selection of the filter capacitor is made on the basis of output ripple specification from Eq. 1.10.

Frequently, however, it is not the output ripple voltage requirement that determines the final choice of the filter capacitor. Ultimately it may be the ripple current rating of the capacitor selected initially on the basis of the output ripple voltage specification.

In a forward-type converter (as in a buck regulator), the capacitor ripple current is greatly limited by the output inductor in series with it (Section 1.3.6). In a flyback, however, the full DC load current flows from common through the capacitor during the transistor "on" time. During the transistor "off" time, a charge of equal ampere-second product must flow into the capacitor to replenish the charge it lost during the "on" time. Assuming, as in Figure 4.1, a sum of "on" time plus reset time of 80% of full period, the RMS ripple current in the capacitor is closely

$$I_{\text{rms}} = I_{\text{dc}}\sqrt{\frac{t_{\text{on}}}{T}} = I_{\text{dc}}\sqrt{0.8} = 0.89 I_{\text{dc}} \tag{4.15}$$

If the capacitor initially selected on the basis of output ripple voltage specifications did not also have the ripple current rating of Eq. 4.15, a larger capacitor or more units in parallel must be chosen.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

4.7 Universal Input Flybacks for 120-V AC Through 220-V AC Operation

Here we consider universal or wide input range flyback topologies that do not have the auto ranging, voltage doubling methods previously described.

In Section 3.2.1, we considered a commonly used scheme that permitted operation from either a 120-V AC or 220-V AC line with minimal changes. As seen in Figure 3.1 at 120 V AC, switch S1 is thrown to the lower position, making the circuit into a voltage doubler that yields a rectified voltage of 336 V. With 220-V AC, S1 is thrown to the upper position and the circuit becomes a full-wave rectifier with $C1$ and $C2$ in series, yielding about 308 V. The converter is thus designed to always work from a rectified nominal input of 308 to 336 V DC by proper choice of the transformer turns ratio.

In some applications, it is preferable to eliminate the requirement of changing $S1$ from one position to the other in changing from 120- to 220-V AC operation. To change switch position without opening the power supply case, the switch must be accessible externally, and this is a safety hazard. The alternative is to change the switch internally, but this requires opening the power supply case to make the change, and this is a nuisance. Further, there is always the possibility that the switch is mistakenly thrown to the voltage doubling position when operated from 200 V AC. This, of course, would cause significant damage—the power transistor, rectifiers, and filter capacitors would be destroyed.

An alternative is the universal line voltage unit that does not require switching and can tolerate the full range of line inputs from 115 to 220 V AC. The rectified 115 V input will be 160 V DC and the 220 V AC will be 310 V DC.

A flyback converter, designed with a small primary/secondary turns ratio, can ensure that the "off"-voltage stress at high AC input does not overstress the power transistor.

The maximum "on" time $\overline{T_{on}}$ at the minimum value of the 220-V AC input is calculated from the corresponding minimum rectified DC input as in Eq. 4.7 and the rest of the magnetics design can proceed as shown in the text following Eq. 4.7. The minimum "on" time occurs at the maximum value of the 220-V AC input. Since the feedback loop keeps the product of $V_{dc}T_{on}$ constant (Eq. 4.3), minimum "on" time is $\underline{T_{on}} = \overline{T_{on}}(\underline{V_{dc}}/\overline{V_{dc}})$ where $\underline{V_{dc}}$ and $\overline{V_{dc}}$ correspond to the minimum and maximum values of the 220-V AC line.

The maximum "on" time with 115-V AC input is still given by Eq. 4.7 and will be greater than with 220 V, as the term $\underline{V_{dc}} - 1$ is smaller. But the primary inductance L_p given by Eq. 4.8, which is proportional to the product $V_{dc}T_{on}$, is still the same as that product is kept constant by the feedback loop. So long as the transistor can operate with

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

the minimum "on" time calculated for the maximum DC corresponding to high AC input, there is no problem. With bipolar transistors operating at a high frequency, transistor storage time could prevent operation at too low an "on" time. An example will clarify this.

Eq. 4.4 gives the maximum "off" stress in terms of the maximum DC input voltage, the output voltage, and the N_p/N_s turns ratio. Assume in that equation that $\overline{V_{\text{ms}}}$ is 500 V; many bipolar transistors can safely sustain that voltage with a negative base bias at turn "off" (V_{cev} rating). At 220 V AC, the nominal V_{dc} is 310 V. Assume that the maximum at high line with a worst-case transient is 375 V. Then for a 5-V output, Eq. 4.4 gives a turns ratio of 21.

Now assume that minimum DC supply voltage is 80% of nominal. Assume a switching frequency of 50 kHz (period T of 20 μs). Maximum "on" time is calculated from Eq. 4.7 at the minimum DC input corresponding to minimum AC input of 0.8 × 115 or 92 V AC. For the corresponding DC input of 1.41 × 92 or about 128 V, maximum "on" time calculated from Eq. 4.7 is 7.96 μs.

Minimum "on" time occurs at maximum input voltage. Assuming a 20% high line, the maximum DC input is $1.2 \times 220 \times 1.41 = 372$ V. Since the feedback loop keeps the product of $V_{\text{dc}}T_{\text{on}}$ constant (Eq. 4.3), "on" time at the 20% high line of 264 V AC is (128/372)(7.96) or 2.74 μs. The circuit can thus cope with either a 20% low AC line input of 92 V AC from a nominal 115 V AC, or a 20% high AC input of 264 V AC from the nominal 220-V AC line by readjusting its "on" time from 7.96 to 2.74 μs.

If this were attempted at higher switching frequencies, the minimum "on" time at a 220-V AC line would become so low as to prohibit the use of bipolar transistors, which could have 0.5- to 1.0-μs storage time. The upper-limit switching frequency at which the above scheme can be used with bipolar transistors is about 100 kHz.

It is instructive to complete the above design. Assume an output power of 150 W at 5-V output. Then $R_o = 0.167\ \Omega$ and the primary inductance from Eq. 4.8 is

$$L_p = \left(\frac{0.167}{2.5 \times 20 \times 10^{-6}}\right)\left(\frac{128 \times 7.96 \times 10^{-6}}{5}\right)^2$$
$$= 139\ \mu\text{H}$$

and the peak primary current from Eq. 4.9 is

$$I_p = \frac{128 \times 7.96 \times 10^{-6}}{139 \times 10^{-6}} = 7.33\ \text{A}$$

There are many reasonably priced bipolar transistors with a V_{cev} rating above 500 V having adequate gain at 7.33 A.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Table 4.1 shows that the 55932 MPP core can tolerate a maximum of 480 ampere-turns, beyond which its inductance will fall off by more than 10% (at 5 A, column 9 shows that the maximum turns is 96 for a maximum inductance of 265 μH). For maximum ampere-turns, the inductance is $32{,}000 \times 0.9(66/1000)^2 = 125\mu$ H. If (as discussed in Section 4.2.3.2) 10% more turns are added, the inductance at 7.33 A will increase by 10% to 138 μH, but at zero current, the inductance will "swing" up to 20% above that.

If the 20% inductance swing is undesirable, the lower permeability core 55933 of Table 4.1 can be used. Table 4.1 shows that the maximum ampere-turns stress is 859. For 7.33 A, the maximum number of turns is 859/7.33 or 117. The maximum inductance for a swing of only 10% is $(0.117)^2 \times 18000 \times 0.9$ or 222 μH. For the desired 139 μh, the required turns are $1000\sqrt{0.139/18 \times 0.95} = 90$.

Thus a design not requiring voltage doubling/full-wave rectifier switching when operation is changed from 115 to 220 V AC is possible. But this subjects the power transistor to a leakage inductance spike at turn "off" of about 500 V. The lower reliability of this scheme must be weighed against the use of a double-ended forward converter or half bridge—both of which subject the "off" transistor to only the maximum DC input (375 V in the preceding example) with no leakage spike. Of course, for 115/220-V AC operation, the rectifier switching of Figure 3.1 must be accepted.

After Pressman *Modern FETs (for example, the Power Integrations "Top Switch" devices) very much simplify the design of universal input flyback type supplies, which are now an accepted and standard topology for lower power applications. Very good application notes are available for these devices. ∼K.B.*

4.8 Design Relations—Continuous-Mode Flybacks

4.8.1 The Relation Between Output Voltage and "On" Time

Look once again at Figure 4.1. When the transistor Q1 is "on," the voltage across the primary is close to V_{dc}–1 with the dot end negative with respect to the no-dot end, and the core is driven—say, up the hysteresis loop. When the transistor turns "off," the magnetizing current reverses the polarity of all voltages in order to remain constant. The primary and secondary are driven positive, but the secondary is caught and clamped to V_{om}+ 1 by $D2$—assuming a 1-V forward drop.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

This reflects across to the primary as a voltage $(N_p/N_s)(V_{\mathrm{om}} + 1)$, with the dot end now positive with respect to the no-dot end. All the current that was flowing in the primary (I_{PO} in Figure 4.2*c*) now transfers to the secondary as I_{TU} in Figure 4.2*d*. The initial magnitude of the secondary current I_{TU} is equal to the final primary current at the end of the "on" time (I_{PO}) times the turns ratio N_p/N_s. Since the dot end of the secondary is now positive with respect to the no-dot end, the secondary current ramps downward with the slope UV in Figure 4.2*d*.

Since the primary is assumed to have zero DC resistance, it cannot sustain a DC voltage averaged over many cycles. Thus in the steady state, the volt-second product across it when the transistor is "on" must equal that across it when the transistor is "off"—i.e., the voltage across the primary averaged over a full cycle must equal zero. This is equivalent to saying the core's downward excursion on the BH loop during the "off" time is exactly equal to the upward excursion during the "on" time. Then

$$(\underline{V_{\mathrm{dc}}} - 1)\overline{t_{\mathrm{on}}} = (V_{\mathrm{om}} + 1)\frac{N_p}{N_s}t_{\mathrm{off}}$$

or

$$V_{\mathrm{om}} = \left[\left(\underline{V_{\mathrm{dc}}} - 1\right)\frac{N_s}{N_p}\frac{\overline{t_{\mathrm{on}}}}{t_{\mathrm{off}}}\right] - 1 \tag{4.16}$$

and since there is no dead time in continuous mode, $\overline{t_{\mathrm{on}}} + t_{\mathrm{off}} = T$, and

$$V_{\mathrm{om}} = \left[\frac{(\underline{V_{\mathrm{dc}}} - 1)(N_s/N_p)(\overline{t_{\mathrm{on}}}/T)}{1 - \overline{t_{\mathrm{on}}}/T}\right] - 1 \tag{4.17a}$$

$$= \left[\frac{(\underline{V_{\mathrm{dc}}} - 1)(N_s/N_p)}{(T/\overline{t_{\mathrm{on}}}) - 1}\right] - 1 \tag{4.17b}$$

The feedback loop regulates against DC input voltage changes by decreasing t_{on} as V_{dc} increases, or increasing t_{on} as V_{dc} decreases.

4.8.2 Input, Output Current–Power Relations

In Figure 4.6, the output power is equal to the output voltage times the average of the secondary current pulses. For I_{csr} equal to the current at the center of the ramp in the secondary current pulse

$$\begin{aligned} P_o &= V_o I_{\mathrm{csr}}\frac{t_{\mathrm{off}}}{T} \\ &= V_o I_{\mathrm{csr}}(1 - \overline{t_{\mathrm{on}}}/T) \end{aligned} \tag{4.18}$$

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

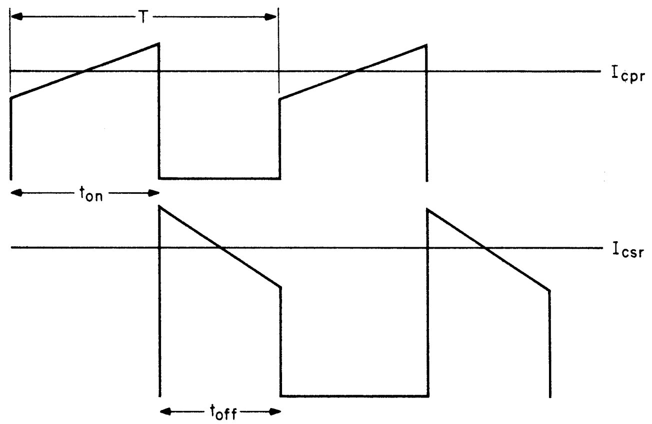


FIGURE 4.6 Real-time relation between the primary and secondary current waveforms in a continuous-mode flyback converter. Current is delivered to the output capacitor only during the "off" period of Q1. At a fixed DC input voltage, t_{on} and t_{off} remain constant. Output load current changes are accommodated by the feedback loop by changing the magnitude of the current at the center of the primary current ramp I_{cpr}, which results in a change at the center of the secondary current ramp (I_{csr}). This occurs over many switching cycles by temporary increases in "on" time until the average current pulse amplitudes build up and then relax to the new steady-state values of t_{on} and t_{off}.

or

$$I_{csr} = \frac{P_o}{V_o(1 - \overline{t_{on}}/T)} \tag{4.19}$$

In Eqs. 4.18 and 4.19, $\overline{t_{on}}/T$ is given by Eq. 4.17 for specified values of V_{om} and V_{dc}, and turns ratio N_s/N_p from Eq. 4.4, which was chosen for acceptably low maximum "off"-voltage stress at maximum DC input.

Further, for an assumed efficiency of 80%, $P_o = 0.8P_{in}$ and I_{cpr} is equal to the current at the center of the ramp in the primary current pulse:

$$P_{in} = 1.25 \quad P_o = V_{dc} I_{cpr} \frac{\overline{t_{on}}}{T}$$

or

$$I_{cpr} = \frac{1.25 P_o}{(V_{dc})(\overline{t_{on}}/T)} \tag{4.20}$$

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

After Pressman *In the continuous mode, the duty cycle is defined by the voltage ratio. Changes in load current try to reflect into the primary, but for transient load changes, the transformer inductance limits the rate of change of current. Hence the first and immediate effect of a transient load increase is to cause a decrease in output voltage, resulting in an increase in the "on" period of Q1 (to increase the primary current). But this results in a further drop in output voltage because there is an immediate decrease in the energy-transferring "off" period (the secondary conducting period). It takes many cycles before the new higher current conditions are established, at which point the duty cycle returns to its original value. This is a dynamic effect intrinsic to the topology and cannot be compensated by the control loop. In terms of control theory, this translates to a right-half-plane-zero. ~K.B.*

4.8.3 Ramp Amplitudes for Continuous Mode at Minimum DC Input

It has been shown that the threshold of continuous-mode operation occurs when there is just the beginning of a step at the front end of the primary current ramp. Referring to Figure 4.6, the step appears when the current at the center of the primary ramp I_{cpr} just exceeds half the ramp amplitude dI_p. That value of $I_{cpr}(\underline{I_{cpr}})$ is then the minimum value at which the circuit is still in the continuous mode. From Eq. 4.20, I_{cpr} is proportional to output power and hence for the minimum output power $\underline{P_o}$ corresponding to $\underline{I_{cpr}}$

$$I_{cpr} = \frac{dI_p}{2} = \frac{1.25\underline{P_o}}{(\underline{V_{dc}})(\overline{t_{on}}/T)}$$

or

$$dI_p = \frac{2.5\overline{P_o}}{(V_{dc})(\overline{t_{on}}/T)} \tag{4.21}$$

In Eq. 4.21, $\overline{t_{on}}$ is taken from Eq. 4.17 at the corresponding value minimum of $V_{dc}(\underline{V_{dc}})$. The slope of the ramp dI_p is given by $dI_p = (\underline{V_{dc}} - 1)\overline{t_{on}}/L_p$, where L_p is the primary magnetizing inductance. Then

$$\begin{aligned} L_p &= \frac{(\underline{V_{dc}} - 1)\overline{t_{on}}}{dI_p} \\ &= \frac{(\underline{V_{dc}} - 1)(\underline{V_{dc}})(\overline{t_{on}})^2}{2.5P_oT} \end{aligned} \tag{4.22}$$

Here again, P_o is the minimum specified value of output power and $\overline{t_{on}}$ is the maximum "on" time calculated from Eq. 4.17 at the minimum specified DC input voltage V_{dc}.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

4.8.4 Discontinuous- and Continuous-Mode Flyback Design Example

It is instructive to compare discontinuous- and continuous-mode flyback designs at the same output power levels and input voltages. The magnitudes of the currents and primary inductances will be revealing.

Assume a 50-W, 5-V output flyback converter operating at 50 kHz from a telephone industry prime power source (38 V DC minimum, 60 V maximum). Assume a minimum output power of one-tenth the nominal, or 5 W.

Consider first a discontinuous-mode flyback. Choosing a bipolar transistor with a 150-V V_{ceo} rating is very conservative, because it is not necessary to rely on the V_{cer} or V_{cev} ratings that permit larger voltages. Then in Eq. 4.4, assume that the maximum "off"-voltage stress $\overline{V_{ms}}$ without a leakage spike is 114 V, which permits a 36-V leakage spike before the V_{ceo} limit is reached. Then Eq. 4.4 gives $N_p/N_s = (114 - 60)/6 = 9$.

Eq. 4.7 gives the maximum "on" time as

$$\overline{t_{on}} = 6 \times 9 \times 0.8 \frac{20 \times 10^{-6}}{37 + 6 \times 9}$$

$$= 9.49\ \mu s$$

and primary inductance for $R_o = 5/10 = 0.5\ \Omega$ from Eq. 4.8 is

$$L_p = \frac{0.5}{2.5 \times 20^{-6}} \left(\frac{38 \times 9.49}{5} \right)^2 \times 10^{-12}$$

$$= 52\ \mu H$$

Peak primary current from Eq. 4.9 is

$$I_p = \frac{38 \times 9.49 \times 10^{-6}}{52 \times 10^{-6}}$$

$$= 6.9\ A$$

and the start of the secondary current triangle is

$$I_{s(peak)} = (N_p/N_s) I_p = 9 \times 6.9 = 62\ A$$

Recall that in the discontinuous flyback, the reset time Tr—the time for the secondary current to decay back to zero—plus the maximum "on" time is equal to $0.8T$ (Eq. 4.6). Reset time is then $T_r = (0.8 \times 20) - 9.49 = 6.5\ \mu s$, and the average value of the secondary current triangle (which

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

should equal the DC output current) is

$$I \text{ (secondary average)} = \frac{I_{\text{s(peak)}}}{2}\frac{T_r}{T}$$
$$= \left(\frac{62}{2}\right)\frac{6.5}{20} = 10 \text{ A}$$

which is the DC output current.

Now consider a continuous-mode flyback for the same frequency, input voltages, output power, output voltage, and the same N_p/N_s ratio of 9. From Eq. 4.17*b*, calculate $\overline{t_{\text{on}}}/T$ for $\underline{V_{\text{dc}}} = 38\ V$ as

$$5 = \left[\frac{(37/9)(\overline{t_{\text{on}}}/T)}{1 - \overline{t_{\text{on}}}/T}\right] - 1$$

or $\overline{t_{\text{on}}}/T = 0.5934$ and $\overline{t_{\text{on}}} = 11.87$ µs, $t_{\text{off}} = 8.13$ µs and from Eq. 4.19

$$I_{\text{csr}} = \frac{50}{(5)(1 - 0.5934)} = 24.59 \text{ A}$$

and the average of the secondary current pulse, which should equal the DC output current, is

$$I\text{(secondary average)} = I_{\text{csr}}(t_{\text{off}}/T) = 24.59 \times 8.13/20 = 10.0 \text{ A}$$

which checks. From Eq. 4.20, $I_{\text{cpr}} = 1.25 \times 50/(38)(11.86/20) = 2.77$ A.

From Eq. 4.22, for the minimum input power of 5 W at the minimum DC input voltage of 38 V, $L_p = 37 \times 38(11.86)^2 \times 10^{-12}/2.5 \times 5 \times 20 \times 10^{-6} = 791$ µH.

The contrast between the discontinuous and continuous modes will now be clear from the following table, which compares the required primary inductances, and primary and secondary currents at minimum DC input of 38 V.

	Discontinuous	Continuous
Primary inductance, µH	52	791
Primary peak current, A	6.9	2.77
Secondary peak current, A	62.0	24.6
On time, µs	9.49	11.86
Off time, µs	6.5	8.13

The lower primary current and especially the secondary current for the continuous mode are certainly an advantage, but the much larger primary inductance that slows up response to load current changes, and the right-half-plane-zero that requires a very low error-amplifier

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

bandwidth to achieve loop stabilization, can make the continuous mode a less desirable choice in applications that require good transient load response. In fixed-load applications this is not a problem.

4.9 Interleaved Flybacks

An interleaved flyback topology is shown in Figure 4.7. It consists of two or more discontinuous-mode flybacks whose power transistors

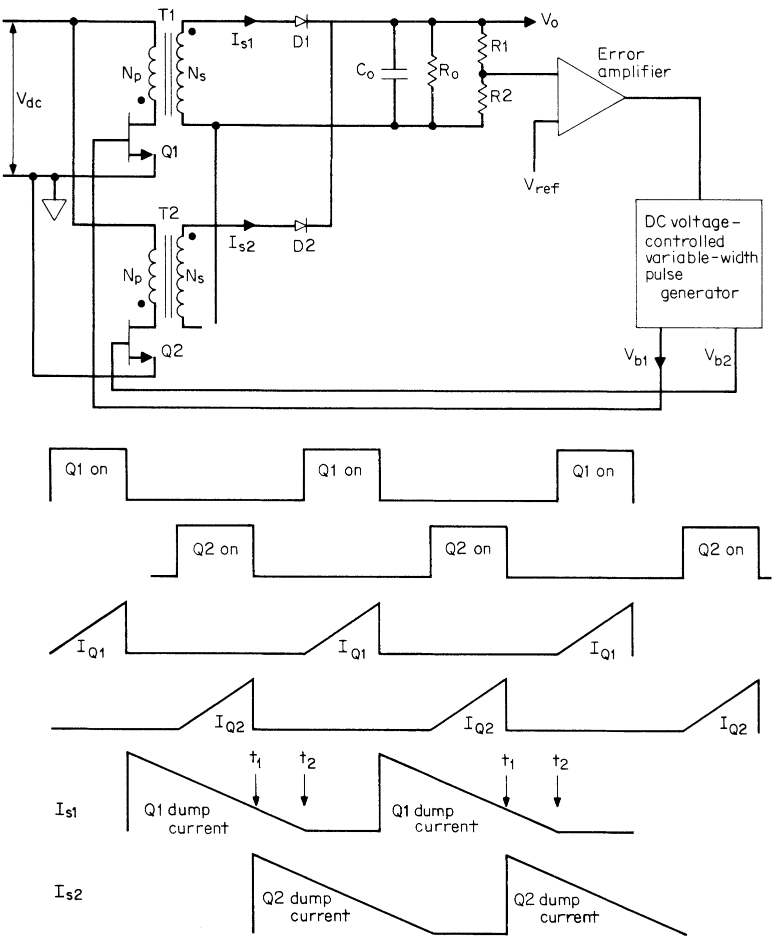


FIGURE 4.7 Interleaving two discontinuous-mode flybacks on alternative half cycles to reduce peak currents. Output powers of up to 300 W are possible with reasonably low peak currents.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

are turned "on" at alternate half cycles and whose secondary currents are summed through their rectifying diodes.

It can be used at power levels up to 300 W, limited mainly by the high-peak primary and especially secondary currents. Although that power level can be obtained with a single continuous-mode flyback with reasonable currents, it may be better to accept the greater cost and volume of two or more interleaved discontinuous-mode flybacks. Both input and output ripple currents are much smaller and of higher frequency. Increasing the number of elements with suitable phase shift between drive pulses will further reduce the ripple current. Further, the discontinuous mode's faster response to load current changes, greater error-amplifier bandwidth, and the elimination of the right-half-plane-zero loop stabilization problem may make this a preferred choice.

A single discontinuous-mode flyback at the 300-W level is impractical because of the very high peak primary and secondary currents, as can be seen from Eqs. 4.2, 4.7, and 4.8.

At a lower power of 150 W, a single forward converter is very likely a better choice than the two interleaved flybacks because of the considerably lower secondary peak current of the forward converter. The interleaved flyback has been shown here for the sake of completeness and for its possible use at lower power levels when many (over five) outputs are required.

4.9.1 Summation of Secondary Currents in Interleaved Flybacks

The magnetics design of each flyback in an interleaved flyback proceeds exactly as for a single flyback at half the power level, because the secondary currents add into the output through their "ORing" rectifier diodes.

Even when both secondary diodes dump current simultaneously (as from t_1 to t_2), there is no possibility that one diode can back-bias the other and supply all the load current. This can happen if one attempts to sum the currents of two low-impedance voltage sources. If one of the low-impedance voltage sources has a slightly higher open-circuit voltage or a lower forward-drop OR diode, it will back-bias the other diode and supply all the load current by itself. This can over-dissipate the diode or the transistor supplying that diode.

Looking back into the secondary of a flyback, however, there is a high-impedance current source, which is the secondary inductance. Thus the current dumped into the common load by either diode is unaffected by the other diode simultaneously supplying load current.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

4.10 Double-Ended (Two Transistor) Discontinuous-Mode Flyback

4.10.1 Area of Application

The topology is shown in Figure 4.8*a*. Its major advantage is that, using the scheme of the double-ended forward converter of Figure 2.13, its power transistors in the "off" state are subjected to only the maximum DC input voltage. This is a significant advantage over the single-ended forward converter of Figure 4.1, where the maximum "off"-voltage stress is the maximum DC input voltage plus the reflected secondary voltage $(N_p/N_s)(V_o + 1)$ plus a leakage inductance spike that may be as high as one-third of the DC input voltage.

4.10.2 Basic Operation

The lower "off"-voltage stress comes about in the same way as for the double-ended forward converter of Figure 2.13. Power transistors $Q1$, $Q2$ are turned "on" simultaneously. When they are "on," the dot end of the secondary is negative, $D3$ is reverse-biased, and no secondary current flows. The primary is then just an inductor, and current in it ramps up linearly at a rate of $dI_1/dt = V_{dc}/(L_m + L_l)$, where L_m and L_l are the primary magnetizing and leakage inductances, respectively. When $Q1$ and $Q2$ turn "off," as in the previous flybacks, all primary and secondary voltages reverse polarity, $D3$ becomes forward-biased, and the stored energy in $L_m = 1/2L_m(I_1)^2$ is delivered to the load.

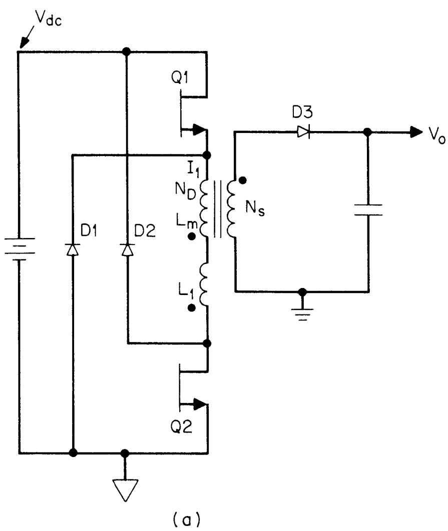
As shown previously, the "on" or set volt-second product across the primary must equal the "off" or reset volt-second product. At the instant of turn "off," the bottom end of L_l attempts to go far positive but is clamped to the positive end of V_{dc}. The top end of L_m attempts to go far negative but is clamped to the negative end of V_{dc}. Thus the maximum voltage stress at either $Q1$ or $Q2$ can never be more than V_{dc}.

The actual resetting voltage V_r across the magnetizing inductance L_m during the "off" time is given by the voltage reflected from the secondary $(N_p/N_s)(V_o + V_{D3})$. The voltage across L_m and L_l in series is the DC supply voltage, and hence, as seen in Figure 4.8*b*, the voltage across the leakage inductance L_l is $V_l = (V_{dc} - V_r)$.

The division of the V_{dc} supply voltage across L_m and L_l in series during the "off" time is a very important point in the circuit design and establishes the transformer turns ratio N_p/N_s as discussed below.

The price paid for this advantage is, of course, the requirement for two transistors and the two clamp diodes, $D1$, $D2$.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.



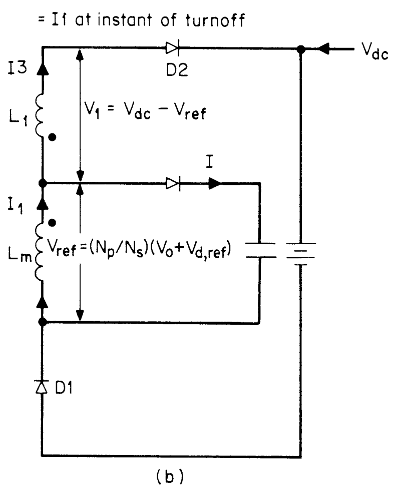


FIGURE 4.8 Circuit during $Q1$ and $Q2$ "off" time. Current I_1, stored in L_m during $Q1$, $Q2$ "on" time, also flows through leakage inductance L_l. During the "off" time, energy stored in L_m must be delivered to the secondary load as reflected into the primary across L_m. But I_1 also flows through L_l, and during the "off" time, the energy it represents ($^1/_2 L_l I^2$) is returned to the input source V_{dc} through diodes $D1$, $D2$. This robs energy that should have been delivered to the output load and continues to rob energy until I_1, the leakage inductance current, falls to zero. To minimize the time for I_1 in L_l to fall to zero, V_l is made significantly large by keeping the reflected voltage $V_r (= N_p/N_s)(V_o + V_{D3})$ low by setting a low N_p/N_s turns ratio. A usual value for V_r is two-thirds of the minimum V_{dc}, leaving one-third for V_l.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

Flyback Converter Topologies

4.10.3 Leakage Inductance Effect in Double-Ended Flyback

Figure 4.8*b* shows the circuit during the $Q1$, $Q2$ "off" time. The voltage across L_m and L_l in series is clamped to V_{dc} through diodes $D1$, $D2$ The voltage V_r across the magnetizing inductance is clamped against the reflected secondary voltage and equals $(N_p/N_s)(Vo + V_{D3})$. The voltage across L_l is then $V_l = V_{dc} - V_r$.

At the instant of turn "off," the same current I_1 flows in L_m and $L_l (I_3 = I_1$ at instant of turn "off"). That current in L_l flows through diodes $D1$, $D2$ and returns its stored energy to the supply source V_{dc}. The L_l current decays at a rate of $dI_1/dt = V_l/L_l$ as shown in Figure 4.9*a* as slope AC or AD. The current in L_m (initially also equal to I_1) decays at a rate V_r/L_m and is shown in Figure 4.9*a* as slope AB.

The current actually delivering power to the load is I_2—the difference between the currents in L_m and L_l. This is shown as current RST in Figure 4.9*b* if the L_1 current slope is AC of Figure 4.9*a*. The larger area current UVW in Figure 4.9*c* results if the L_1 current slope is faster, as AD of Figure 4.9*a*. It should be evident in Figures 4.9*b* and 4.9*c* that so long as current still flows in leakage inductance L_l, through $D1$ and $D2$ back into the supply source, all the current available in L_m does

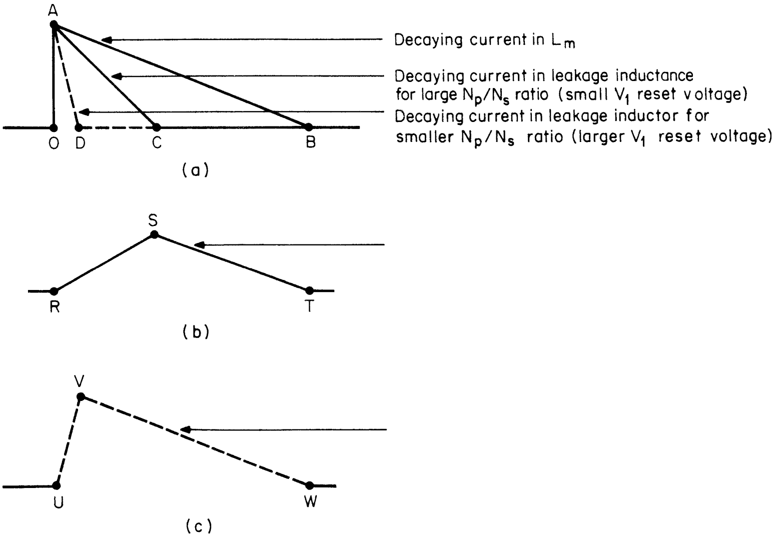


FIGURE 4.9 (*a*) Currents in magnetizing and leakage inductances in double-ended flyback. (*b*) Current into reflected load impedance for large N_p/N_s ratio. AB–AC of Figure 4.9*a*. (*c*) Current into reflected load impedance for smaller N_p/N_s ratio. AB–AD of Figure 4.9*a*.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.

not flow into the reflected load but is partly diverted back into the supply.

It can thus be seen from Figures 4.9*b* and 4.9*c* that to maximize the transfer of L_m current to the reflected load and to avoid a delay in the transfer of current to the load, the slope of the leakage inductance current decay should be maximized (slope *AD* rather than *AC* in Figure 4.9*a*). Or in magnetics–power supply jargon, the leakage inductance current should be rapidly reset to zero.

Since the rate of decay of the leakage inductance current is V_l/L_l and $V_l = V_{dc} - (N_p/N_s)(V_o + V_{D3})$, choosing lower values of N_p/N_s increases V_l and hastens leakage current reset. A usual value for the reflected voltage $(N_p/N_s)(V_o + V_{D3})$ is two-thirds of V_{dc}, leaving one-third for V_l. Too low a value for V_r will require a longer time to reset the magnetizing inductance, rob from the available *Q*1, *Q*2 "on" time, and decrease the available output power.

Once N_p/N_s has been fixed to yield $V_l = V_{dc}/3$, the maximum "on" time for discontinuous operation is calculated from Eq. 4.7, L_m is calculated from Eq. 4.8 and I_p from Eq. 4.9, just as for the single-ended flyback.

References

1. Billings, K., *Switchmode Power Supply Handbook,* McGraw-Hill, New York, 1989.
2. Chryssis, G., *High Frequency Switching Power Supplies,* 2nd Ed., McGraw-Hill, New York, 1989, pp. 122–131.
3. Dixon, L., "The Effects of Leakage Inductance on Multi-output Flyback Circuits," *Unitrode Power Supply Design Seminar Handbook,* Unitrode Corp., Lexington, Mass., 1988.
4. Patel, R., D. Reilly, and R. Adair, "150 Watt Flyback Regulator," *Unitrode Power Supply Design Seminar Handbook,* Unitrode Corp., Lexington, Mass., 1988.

Downloaded from Digital Engineering Library @ McGraw-Hill (www.digitalengineeringlibrary.com)
Copyright © 2009 The McGraw-Hill Companies. All rights reserved.
Any use is subject to the Terms of Use as given at the website.